Floor Plan Aware Feature Engineering and Gradient Boosted Learning for BLE Based Indoor Localization

SurajPaliwal[1],*Shreya Pawalia[2], Tomohiro Shibata[3 †]

[1,3,]Kyushu Institute of Technology, Kitakyushu, Japan

[2]Dronacharya College of Engineering, Gurugram, Haryana

Abstract

In nursing care facilities, practical challenges with Bluetooth Low Energy (BLE)-based indoor localization include extreme RSSI noise, temporal sparsity, and class imbalance between rooms. This study does not introduce a new localization algorithm; instead, it proposes a floor plan aware feature engineering framework that transforms raw RSSI signals into spatially structured representations by embedding prior knowledge of beacon placement, room adjacency, and temporal activity patterns. RSSI statistics, zone based aggregation, inter zone interactions, proximitybased transformations, and temporal behavioral indicators that represent routine movement dynamics inside the facility are all included in the designed features. XGBoost obtains 73.35% accuracy under stratified evaluation (weighted F1: 0.7282). The accuracy of Leave-One-Day-Out cross validation (LODO-CV), which is used to evaluate resilience against temporal bias, is 54.1% (weighted F1: 0.5230) under day level generalization. While temporal indicators provide moderate increases under rigorous validation, ablation analysis shows that spatial feature groupings contribute cooperatively. The majority of errors, according to confusion matrix analysis, happen between geographically neighboring rooms, indicating coherent spatial learning rather than random signal correlation. These findings show that feature design that incorporates floor plan and behavioral context enhances resilience under practical deployment constraints without the need for more sensors or calibration.

*paliwal.suraj6482@mail.kyutech.jp
†tom@brain.kyutech.ac.jp

1 Introduction

Indoor localization is an essential part of context-aware systems in public buildings, offices, nursing homes, and healthcare facilities [7, 9]. Applications like activity recognition, workflow analysis, safety monitoring, and intelligent resource allocation are made possible by accurate room level localization. Bluetooth Low Energy (BLE) beacons are one of the most popular sensing technologies accessible because of their low power consumption, inexpensive deployment, and connectivity with common mobile devices. [5, 8, 14, 16]

In real-world scenarios, BLE based indoor localization is still difficult. Multipath propagation, human body attenuation, device heterogeneity, and ambient dynamics all contribute to the intrinsic noiseness of Received Signal Strength Indicator (RSSI) measurements [7, 11, 15]. Furthermore, there is frequently a significant class imbalance in operational datasets, with high traffic regions predominating and individual rooms having comparatively few samples [1,2,6]. The generalization performance of traditional fingerprinting techniques that depend on absolute RSSI values is severely constrained by these reasons.

These constraints are explicitly reflected in the Decode the Invisible: Location Recognition Challenge in Care Facilities [17], to which this work is submitted. The challenge calls for room level localization using just the given data, with no extra sensors or external calibration, and presents a BLE dataset that was gathered in a real care facility under realistic deployment circumstances. The necessity for organized and reliable feature representations is shown by the fact that models trained directly on raw RSSI fingerprints under such constraints frequently overfit majority locations and perform poorly on minority classes.

We concentrate on representation design instead of introducing a novel localization algorithm. We present a floor plan aware spatiotemporal feature engineering framework that directly incorporates interior spatial structure and beacon placement knowledge into the feature space. Signal intensity statistics, zone based aggregations aligned with the floor plan, prominent signal regions, relative inter zone relationships, proximity based transformations, and coarse temporal context are among the spatially significant descriptors derived from raw RSSI readings [3, 4]. The methodology emphasizes relative signal behavior rather than depending only on absolute RSSI magnitudes [10], and attempts to increase robustness under RSSI noise and class imbalance by modeling both spatial structure and short term temporal dynamics.

We use only the challenge provided dataset to benchmark representative tree based models and a multilayer perceptron in order to assess the suggested representation. The optimal balance between accuracy and robustness is achieved by gradient boosted decision trees, especially XGBoost, according to experimental results. Furthermore, confusion matrix analysis shows that the majority of misclassifications happen between rooms that are geographically close to one another, suggesting that the learned representations maintain significant spatial coherence.

The main contributions of this work are

- A floor plan-aware spatiotemporal feature engineering framework for BLE based indoor localization.
- Demonstrated robustness under severe RSSI noise and class imbalance without additional sensors or data augmentation.
- Ablation and spatial error analyses revealing complementary roles of temporal continuity and floor plan aware modeling.

2 Related Work

Indoor localization has been intensively investigated using various technologies such as Wi-Fi, RFID, ultra wideband, inertial sensors, and Bluetooth Low Energy (BLE) [7, 9]. Among these, BLE gained special attention because of its low implementation costs and energy economy, which make it appropriate for long term monitoring in settings like hospitals and nursing homes. Conventional BLE localization usually uses fingerprinting, which involves gathering a reference database of RSSI measurements at predetermined sites and comparing them to online observations. However, fingerprinting techniques' resilience is limited in practice because of their high sensitivity to signal fluctuations, environmental changes, and device variability [5, 11, 14, 15].

In order to overcome these constraints, machine learning techniques have gained widespread acceptance. Because tree based classifiers, like Random Forest, can manage missing values and nonlinear feature interactions, they have demonstrated good performance in noisy indoor environments [12]. Additionally, deep learning models have been investigated, especially in the presence of huge labeled datasets [13, 16]. However, deep models frequently need careful adjustment and can overfit when data is sparse or unbalanced.

The importance of feature design and data balancing techniques in the context of class imbalance has been highlighted in recent research. Indoor localization has made use of oversampling techniques like SMOTE and ADASYN; however, synthetic samples can skew underlying signal distributions, particularly when RSSI variance is high [10, 16]. On the other hand, signal pattern based relabeling techniques exhibit enhanced robustness in nursing care settings [1, 2, 10] by utilizing statistical similarities between rooms to support minority classes using real beacon data. In addition, a number of studies emphasize the advantages of simulating geographical context and relative signal correlations. It has been demonstrated that zone based aggregation, dominant beacon analysis, and relative RSSI comparisons enhance generalization without the need for further sensors or calibration.

Our study builds on these insights by combining floor plan aware feature engineering with gradient boosted learning. To achieve strong localization under the realistic restrictions of the ABC 2026 Challenge, we directly encode spatial structure into the feature representation instead of depending solely on synthetic augmentation or raw RSSI fingerprints.

3 Problem Formulation

Let $\mathcal{B} = \{1, 2, \ldots, 25\}$ denote the set of Bluetooth Low Energy (BLE) beacons deployed on the target floor. At each timestamp t, the receiver observes a vector of received signal strength measurements,

$$\mathbf{r}_t = [r_{1,t}, r_{2,t}, \ldots, r_{|\mathcal{B}|,t}], \tag{1}$$

where $r_{i,t}$ denotes the Received Signal Strength Indicator (RSSI) value (in dBm) from beacon $i \in \mathcal{B}$. Valid measurements satisfy $r_{i,t} < 0$, while undetected signals are treated as invalid.

Each timestamp is associated with a location label

$$y_t \in \mathcal{Y}, \tag{2}$$

where $\mathcal{Y}$ denotes the set of indoor locations defined in the ABC 2026 Indoor Localization Challenge [17]. In this study, $|\mathcal{Y}| = 22$, including individual rooms and shared functional areas.

The localization task is to learn a mapping

$$f : \mathbf{x}_t \to y_t, \tag{3}$$

where $\mathbf{x}_t$ represents a feature vector derived from $\mathbf{r}_t$. Given a new observation, the trained model predicts the most probable location $\hat{y}_t$.

In practice, practical deployment limits make this issue more difficult [17]. Multipath propagation, body attenuation, and ambient change all contribute to the inherent noise in RSSI measurements. Additionally, there is a significant class imbalance in the dataset, with many low sample rooms being dominated by a few high traffic sections. Furthermore, there are no auxiliary sensors or external calibration data available. In these circumstances, feature representation becomes just as important as choosing a model. Instead of depending only on raw RSSI fingerprints, effective localization needs features that encode spatial structure, relative connections, and signal statistics in a way that is resistant to imbalance and noise.

3.1 On the Use of Beacon Power Information

Beacon transmission power (Tx-power) can be deduced from the challenge-provided beacon log's *power* column. Inspection, however, shows that this characteristic is not physically interpretable for feature construction, nor is it consistent. The value -2147483648, which is frequently used as a placeholder for missing data and corresponds to the minimum 32-bit signed integer, is present in a significant percentage of entries. The remaining numbers, such as 3, 4, 7, 8, and 12, do not match the normal BLE Tx-power levels in dBm, which typically fall between around −30 and +4 dBm. The *power* attribute is only used during initial data loading and is not included in feature engineering or model training because of the ubiquity of placeholder values and its unclear physical meaning. Therefore, only RSSI measurements and their statistical summaries are used to determine all localization features. This decision is consistent with earlier BLE-based indoor localization research that showed good results using RSSI-only representations and no explicit Tx-power calibration [14–16].

4 Method for Floor Plan Aware BLE Localization

This section explains the suggested indoor localization technique, which blends supervised learning with floor-plan-aware feature engineering.

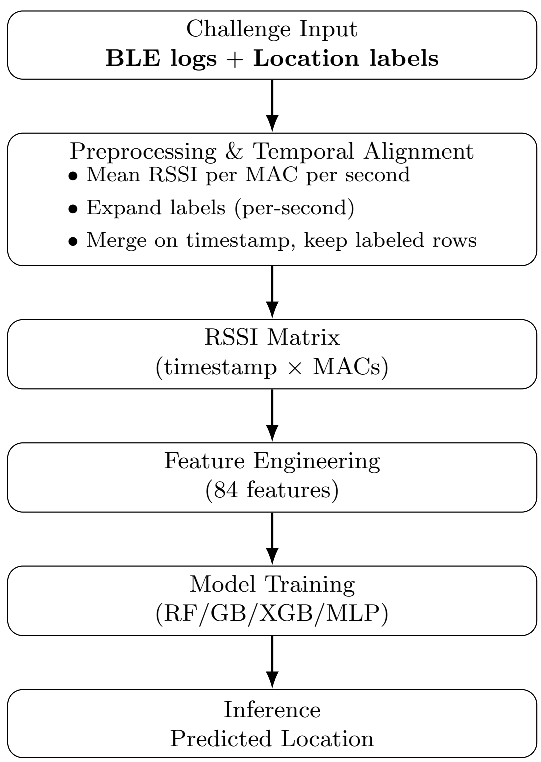


Figure 1: The proposed BLE-based indoor localization framework's overall pipeline includes preprocessing and temporal alignment, as well as feature engineering and supervised learning.

4.1 Preprocessing and Temporal Alignment

The raw BLE dataset consists of scan level records indexed by timestamp, where each entry includes a beacon MAC address, the corresponding RSSI value, and metadata. The ABC 2026 Challenge protocol, as outlined in [1,2,17], is followed by the official relabeling and annotating procedures.

The relabeling rules and location annotations that are provided are not altered in this investigation. Multiple RSSI readings for the same beacon may occur within brief periods due to the 10 Hz operating frequency of BLE beacons. By calculating the mean RSSI per detected beacon, RSSI values are aggregated at the timestamp level to produce a consistent signal representation appropriate for feature extraction. After that, the data is transformed into an RSSI matrix, with each column denoting a distinct beacon MAC address and each row representing a timestamp. In order to maintain a fixed-dimensional representation, missing detections are represented as zero values.

Initially given as time intervals, location annotations are enlarged to second-level timestamps and combined with the aggregated RSSI matrix. For the purpose of supervised learning, only timestamps with labels are kept. Figure 2 shows how the timestamp-level aggregation technique used in this study differs from pivot-based reshaping. The proposed restructuring constructs a compact, fixed-dimensional RSSI matrix, improving numerical stability and computational efficiency without altering the official relabeling protocol.

(a) Pivot-based per-MAC columns

timestamp	mac_addr	RSSI_1	RSSI_2	RSSI_3	RSSI_4	...
13:00:00	4	0	0	0	-91	0
13:00:00	4	0	0	0	-92	0
13:00:00	4	0	0	0	-91	0

(b) Aggregated RSSI matrix constructed in this study

timestamp	mac_1	mac_2	mac_3	mac_4
13:00:00	0	-92	0	-91
13:00:01	0	-95	0	-93

Figure 2: Comparison of BLE log reshaping approaches. (a) Pivot-based reshaping can produce repeated timestamp entries with sparse per-MAC columns. (b) Timestamp-level aggregation used in this study constructs a compact RSSI matrix by averaging RSSI values per beacon within each timestamp.

4.2 Feature Engineering

The RSSI matrix at the timestamp level is converted into a compact feature vector that incorporates proximity encoding, relational descriptors, floor-plan-aware spatial aggregation, statistical summaries, and temporal context to enhance the robustness of localization.

There are three concepts that guide feature design:

- Reduce sensitivity to instantaneous RSSI fluctuations
- Embed structural spatial information obtained from the floor plan
- Preserve relative signal relationships that remain stable under device and environmental variability.

Raw RSSI magnitudes do not always accurately reflect actual distance due to multipath propagation, human body attenuation, and environmental dynamics. The suggested representation focuses structurally consistent signal behavior through aggregation and related encoding rather than simulating fleeting signal spikes. While this assumption may be less effective under extremely sparse beacon deployment or irregular layouts, it provides a stable inductive bias under realistic BLE installations.

4.2.1 Basic Signal Statistics

Global signal statistics for each timestamp are calculated based on authentic RSSI data. These consist of the total signal range, mean RSSI, standard deviation, minimum and maximum values, and the number of beacons detected.

Spatial visibility and beacon density, which vary regularly between enclosed rooms and shared open spaces, are reflected in the number of detections. While dispersion-based measurements encode variability brought on by spatial confinement and obstruction effects, the mean RSSI captures overall signal strength.

These descriptors increase generalization under severe room-level class imbalance and lessen overfitting to unstable fingerprints by summarizing signal behavior without depending on individual beacon serial numbers.

4.2.2 Floor-Plan-Aware Zone Features

BLE beacons are arranged into spatial zones based on where they are physically located within the building in order to explicitly incorporate architectural structure into the portrayal. The floor plan is divided into architecturally functional areas by manually determined bounding boxes, as seen in Fig. 3. The same zone is assigned to beacons that are situated inside the same structural region.

Detection counts and aggregated signal data are calculated for each zone. The idea that structural proximity is more stable than immediate individual RSSI magnitudes is reflected in zone-level pooling. Consistent geographical patterns are highlighted and sensitivity to individual beacon noise is decreased by aggregating signals within spatially adjacent regions.

In contrast to the relabeling algorithms suggested in [1, 2], which refine annotations at the label level, the current method adds structural priors directly into

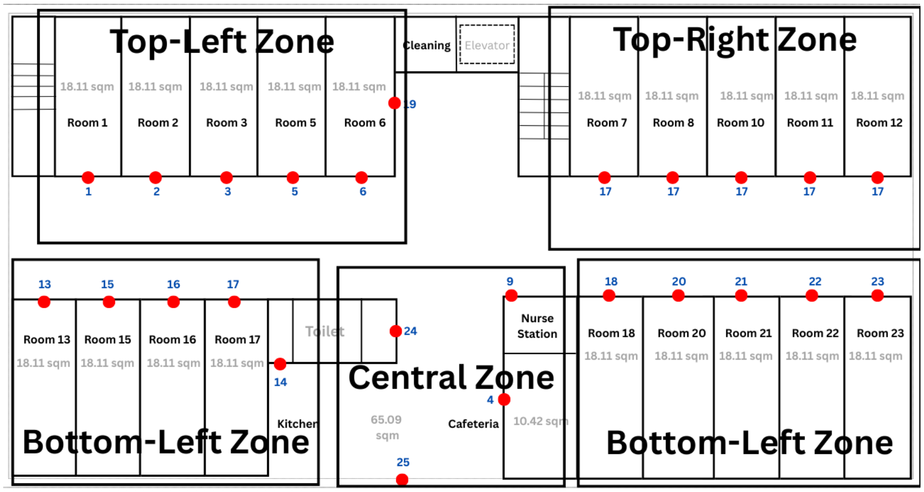


Figure 3: Floor plan showing beacon placement and user-defined zone bounding boxes used for spatial aggregation.

the feature space while maintaining the original ground-truth labels. Therefore, zoning encodes spatial adjacency at the representation level without changing supervision.

Because zone definitions depend on architectural layout and beacon placement density, careful alignment with physical structure is required. While deployment in other facilities would require customization, this design provides a practical and effective mechanism for embedding floor-plan knowledge into learning.

4.2.3 Dominant Zone and Relative Relationships

The dominant zone is defined as the spatial region exhibiting the strongest aggregated signal at a given timestamp. Formally, let $\mu_t^{(k)}$ denote the average RSSI of zone k at time t. The dominant zone is then determined as

$$Z_t^* = \arg\max_k \mu_t^{(k)}. \tag{4}$$

This feature provides a coarse but robust estimate of the most probable structural location by emphasizing stable regional signal behavior rather than transient RSSI spikes. To further capture spatial structure, relative comparisons between neighboring zones are computed. In particular, inter-zone signal ratios are defined as

$$R_{TB}(t) = \frac{\mu_t^{(\text{top-left})}}{\mu_t^{(\text{bottom-left})}}, \tag{5}$$

$$R_{LR}(t) = \frac{\mu_t^{(\text{top-left})}}{\mu_t^{(\text{top-right})}}. \tag{6}$$

Modeling relative differences instead of absolute magnitudes mitigates the impact of global RSSI shifts caused by device heterogeneity or environmental variation. Even when absolute signal strengths fluctuate across days or devices, proportional differences between spatial regions tend to remain more stable.

Together, dominant zone identification and inter-zone relationships encode directional and structural context in a compact and computationally efficient manner, strengthening spatial coherence in predictions.

4.2.4 Top-N and Proximity Features

To preserve fine-grained spatial cues, the three strongest RSSI values at each timestamp are retained. The choice of $N = 3$ balances discriminative capacity and robustness. Including additional weaker signals tends to introduce noise from distant or unstable beacons, whereas smaller values may discard informative proximity cues.

RSSI values are also transformed into non-negative proximity scores using a linear offset. This transformation preserves ordinal relationships without imposing a physical distance interpretation. Rather than estimating metric distance, it provides a monotonic and numerically stable representation of signal closeness that is well-suited for tree-based learning models.

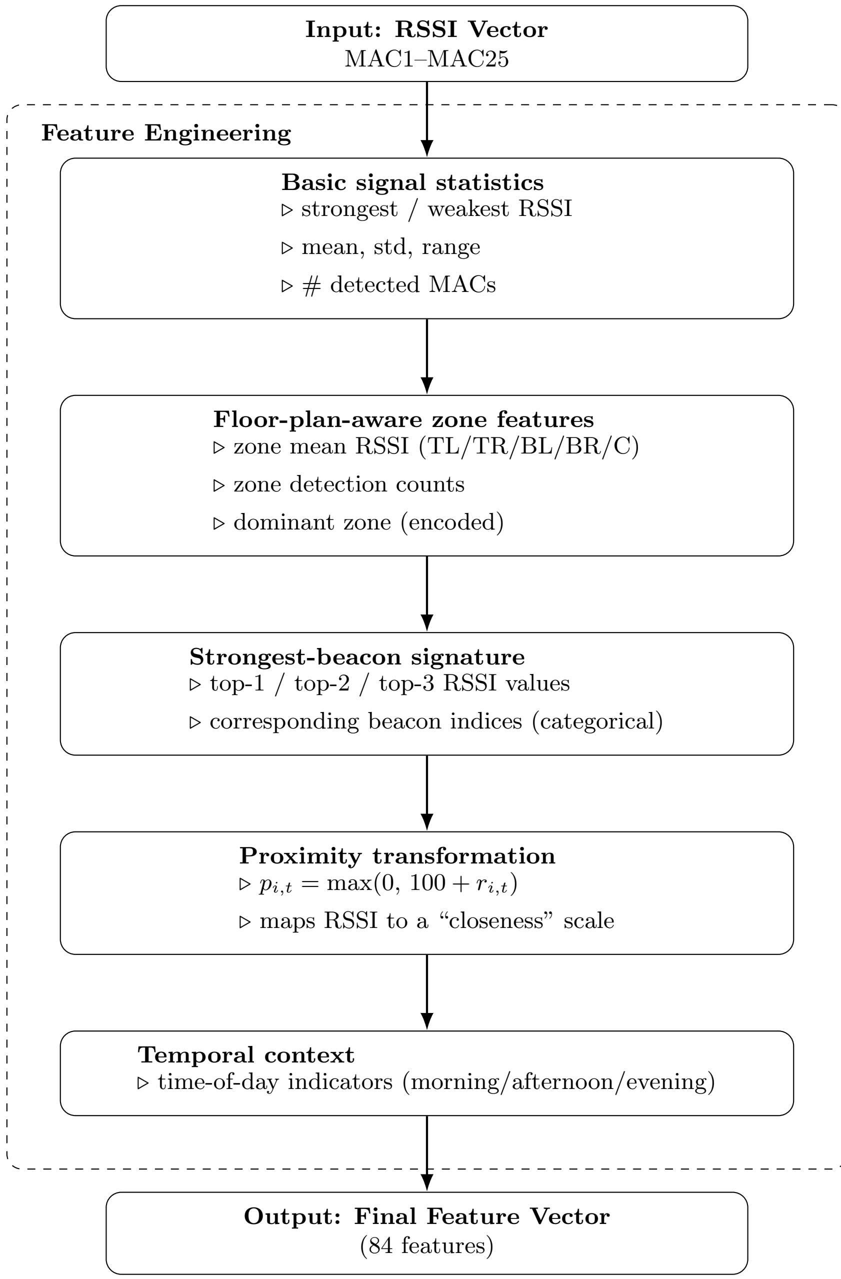
4.2.5 Temporal Context Features

Temporal indicators are incorporated to capture structured behavioral routines within the care facility, such as scheduled visits and medication rounds. Without explicit sequence modeling, coarse time-of-day features encode recurring activity patterns that are not directly observable from RSSI signals alone.

Temporal features are treated as contextual cues rather than primary predictors. This design ensures that spatial signal learning remains dominant while allowing the model to exploit consistent behavioral regularities when present. Since operational schedules differ across facilities, these descriptors are interpreted as environment-dependent contextual signals rather than universally transferable features.

4.3 Model Training

The final feature vector $\mathbf{x}_t$ is generated by concatenating all designed features. In label-encoded categorical variables, missing values are replaced by zeros. The dataset is split into training and test sets using stratified sampling to preserve class distribution. Four supervised models are evaluated: Random Forest, Gradient Boosting, XGBoost, and a multilayer perceptron (MLP). While neural networks are trained on standardized inputs, tree-based models use raw attributes. Hyperparameters are empirically chosen to achieve a balance between generalization performance and model complexity.



Note: Power is aggregated during preprocessing and merged by timestamp.

Figure 4: Feature engineering pipeline. Raw RSSI measurements are transformed into statistical, zone-aware, signature, proximity, and temporal descriptors.

5 Experimental Results and Analysis

The experimental technique complies with the evaluation process outlined in [2] for Decode the Invisible: Location Recognition Challenge in Care Facilities. The main performance indicator used to account for class imbalance across indoor sites is the weighted F1 score. All experiments rely solely on the provided labeled dataset and have no other data sources. The dataset comprises 23,879 samples from 22 indoor sites following preprocessing and feature extraction. To maintain the original class distribution, stratified sampling is used to separate the data into training (80%) and test (20%) groups.

A multilayer perceptron (MLP), Random Forest (RF), Gradient Boosting (GB), and XGBoost (XGB) are the four supervised models that are assessed. In contrast to the MLP, which uses z score normalized inputs, tree based models are trained on raw feature values. In order to balance generalization performance and model capacity, hyperparameters are empirically defined .Model robustness was examined along three complementary axes: (i) class-imbalance aware stratified evaluation, (ii) strict day-level temporal isolation via LODO-CV (AppendixB), and (iii) validation–test consistency analysis to assess overfitting (AppendixB.1.1).

5.1 Overall Model Performance

Table 1 summarizes each evaluated model's performance on the test set. Detailed dataset distribution and room-level imbalance statistics are provided in AppendixA. In terms of overall accuracy (73.35%), weighted F1 score (0.7282), and macro F1 score (0.5957), XGBoost fared better than the other two classifiers. A lower macro F1 indicated that the multilayer perceptron was less robust to minority classes, though having a similar accuracy. Random Forest and Gradient Boosting had strong training accuracy despite showing reduced generalization on the test set, suggesting minor overfitting. In contrast, XGBoost maintained a strong balance between bias and variance by using gradient boosted tree ensembles with regularization. Hyperparameters were optimized using Optuna with inner cross-validation (Appendix B).

Table 1: Model Performance Comparison

Model	**Accuracy**	**F1 (Micro)**	**F1 (Macro)**	**F1 (Weighted)**
Random Forest	0.7016	0.7016	0.5644	0.7004
Gradient Boosting	0.7073	0.7073	0.5345	0.7223
Neural Network (MLP)	0.7282	0.7282	0.4927	0.7028
XGBoost	**0.7335**	**0.7335**	**0.5957**	**0.7282**

5.2 Confusion Matrix Analysis

The row normalized *zone level* confusion matrix produced by the XGBoost model is shown in Figure 5. To enhance interpretability in the face of extreme room level class imbalance, locations were categorized into five predetermined spatial zones according to the floor design. In every zone, the model exhibits high diagonal dominance, suggesting accurate coarse grained localization. Because it has more samples and extensive beacon coverage, the *Central zone* has the highest recall (86.2%). With recalls of 78.2%, 75.2%, and 73.6%, respectively, the *Top Right*, *Bottom Left*, and *Bottom Right* zones likewise demonstrate reliable performance. Due to its proximity to shared corridor areas and few training samples, the *Top Left zone* has a slightly lower recall (58.1%). The majority of misclassifications occur between spatially nearby zones, especially between the Central zone and the periphery zones, suggesting that the errors are spatially coherent rather than the result of random confusion. These findings imply that robust zone level localization is made possible by the suggested floor plan aware feature engineering, which enhances the more detailed room level analysis offered in the supplementary material.

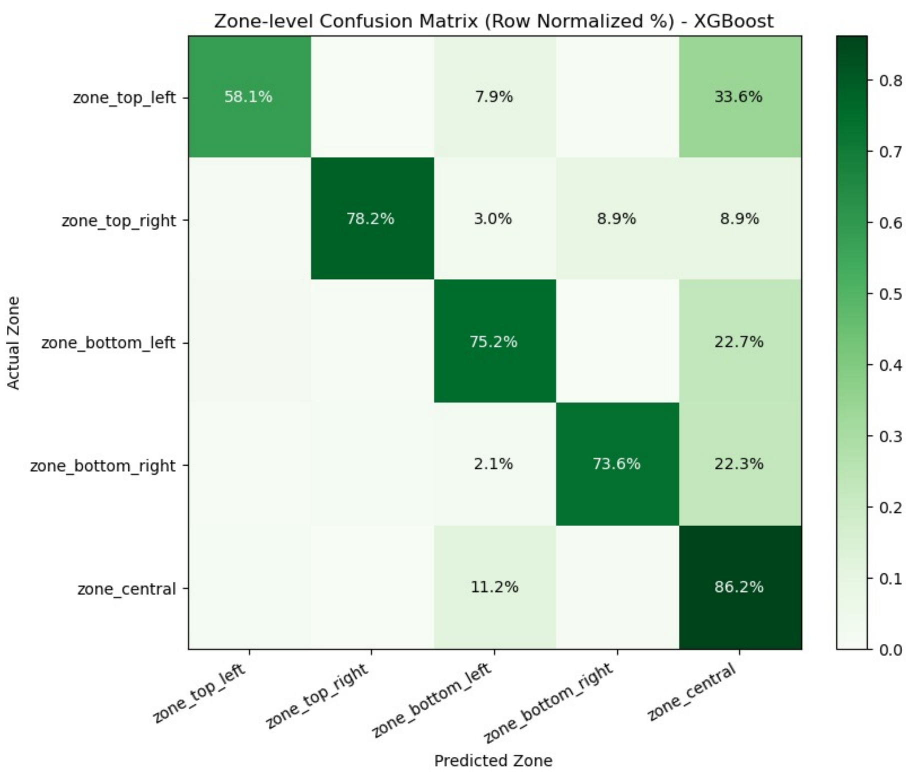


Figure 5: Row normalized zone level confusion matrix for nurse localization using the XGBoost model.

5.3 Feature Group Ablation Analysis

Feature group ablation was used to determine the marginal contribution of each feature family. In each experiment, one feature group is removed from the entire 84 feature configuration in each experiment, but all other elements remain unchanged. In order to evaluate resilience under day level distribution shifts, stratified sampling (with the same split and XGBoost hyperparameters as the main experiment) and Leave One Day Out cross validation (LODO-CV) were both used. Removing temporal indicators under stratified sampling significantly reduces performance (Accuracy: $0.732 \rightarrow 0.629$; Weighted F1: $0.728 \rightarrow 0.613$), suggesting that temporal context reflects structured behavioral regularities in the facility. On the other hand, the room hour distribution analysis (Appendix A) indicates partial temporal label coupling under random splits, with some rooms showing substantial temporal concentration.

To assess generalization, LODO-CV was used with a one day testing window. The marginal contribution of temporal markers becomes moderate (ΔWeighted F1 ≈ 0.01), despite the fact that day level shifts negatively impact overall performance. This implies that while temporal features offer a small but steady benefit under true day separation, the significant increase under stratified sampling could be partially due to temporal overlap.

However, there are only slight differences between the two procedures when certain spatial feature groups zone aggregation, inter zone ratios, proximity transform, strongest beacon signature, and location specific signals are eliminated (Table 2). This suggests that robust cooperative contributions result from the encoding of spatial information using complementary and partially correlated representations. Overall, generalization robustness is improved by floorplan aware spatial encoding, and under more stringent validation, temporal indicators boost performance without overpowering predictions.

Table 2: Feature group ablation results. Each row removes one feature family from the full 84 feature model.

Configuration	**Feat.**	**Accuracy**	**F1 (Weighted)**	ΔWtd
Baseline (All Features)	84	0.7324	0.7277	0.0000
Without Raw MAC RSSI	59	0.7312	0.7263	-0.0014
Without Basic RSSI Statistics	78	0.7347	0.7299	+0.0022
Without Zone based Features	74	0.7330	0.7280	+0.0004
Without Inter zone Ratios	82	0.7314	0.7264	-0.0013
Without Proximity Transform	59	0.7345	0.7297	+0.0020
Without Strongest beacon Signature	78	0.7335	0.7282	+0.0005
Without Location specific Signals	78	0.7339	0.7289	+0.0012
Without Temporal Indicators	80	0.6294	0.6125	-0.1152

6 Discussion

This study illustrates that domain-informed feature engineering can enhance BLE-based indoor localization within realistic deployment restrictions without the necessity of a new localization method. The suggested architecture incorporates floor-plan knowledge directly into the representation through zone aggregation, inter-zone interactions, and proximity-based descriptors, encouraging the model to learn spatially structured patterns instead of depending just on unstable absolute RSSI magnitudes.

This paper makes a practical rather than a methodological contribution. Prior studies have investigated individual components like zone-based aggregation and gradient-boosted trees. To achieve strong performance under the limitations of the *ALRC: Decode the Invisible: Activity and Location Recognition Challenge in Care Facility*, this study methodically combines these components into an integrated feature design that specifically takes architectural structure into account.

The significance of validation design is demonstrated by the contrast between stratified evaluation and Leave-One-Day-Out cross-validation (LODO-CV). Stratified splits allow for some temporal overlap between training and testing data, but they also report higher accuracy. Overall performance declines under rigorous day-level separation, reflecting actual distribution alterations. However, spatial feature groups continue to contribute consistently, suggesting that the representation captures structural regularities that are beyond particular patterns at the day-level.

The role of temporal indicators is complementary. They offer noticeable performance improvements under stratified sampling, partially reflecting the dataset's structured hour-level activity concentration. Their contribution is still positive but more moderate under LODO-CV, indicating that temporal cues provide contextual support instead of serving as the primary predictive drivers.

The majority of misclassifications happen between physically adjacent zones, according to the confusion matrix study. Instead of following random distribution, errors follow floor-plan proximity, suggesting that the learnt representation respects architectural structure. The model seems to internalize spatial-behavioral regularities that are consistent with travel patterns across care facilities, even in the absence of explicit transition modeling or graph-based limitations.

This interpretation is consistent with other research showing that spatial aggregation and relative signal relationships can improve BLE localization robustness without the need for further sensors or calibration. Similar findings have been documented in structured healthcare settings, where spatial priors enhance stability under noisy RSSI.

The current research does not take advantage of temporal sequence modeling, which could further enhance localization stability, and instead concentrates on single-timestep categorization. Furthermore, since the goal was to assess representation resilience under the original data distribution, neither data augmentation nor class rebalancing techniques were used.

Overall, the findings imply that durable structural support under temporal separation can be obtained by directly incorporating architectural context into feature space. Floor-plan-aware representation provides a useful and efficient route toward reliable BLE-based indoor localization within the parameters of the ALRC challenge setting [17].

7 Conclusion

This paper presented a floor-plan-aware feature engineering framework for BLE-based indoor localization under the constraints of the ALRC ABC 2026 Challenge. By incorporating architectural structure into spatial and relational signal descriptors in conjunction with gradient-boosted models, the work concentrated on representation design rather than presenting a novel learning architecture.

Through proximity-based modifications, zone-level aggregation, and inter-zone interactions, the suggested representation increases robustness when compared to raw RSSI inputs. The majority of errors, according to confusion matrix analysis, happen between adjacent rooms, indicating that predictions are more consistent with floor-plan topology than with random signal correlations. The findings also suggest that stable performance is facilitated by spatial structure in conjunction with coarse temporal context.

Although temporal distribution variations cause overall accuracy to decline under strict Leave-One-Day-Out cross-validation, spatial feature contributions stay constant, suggesting robustness beyond simple temporal overlap.

Overall, without the need for further sensors or calibration, integrating architectural priors straight into feature space offers a practical and computationally effective method for BLE-based indoor localization in healthcare settings. Graph-based transition constraints, explicit spatiotemporal modeling, and multi-facility generalization will all be examined in future research.

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Appendix

A Dataset Distribution and Day-Level Robustness Evaluation

A.1 Dataset Overview and Class Imbalance

The dataset includes 23,879 labeled samples with 55 raw attributes before engineering feature augmentation, following preprocessing and feature extraction. There are 84 derived features in the final supervised learning representation. The dataset has 23,879 labeled samples after preprocessing. The room level sample distribution is shown in Table 3.

Table 3: Room level sample distribution (23,879 total samples).

Room	Count	%	Room	Count	%
nurse station	9439	39.5	511	131	0.5
kitchen	5285	22.1	502	78	0.3
cafeteria	4929	20.6	501	64	0.3
hallway	952	4.0	508	52	0.2
cleaning	726	3.0	503	44	0.2
523	420	1.8	510	35	0.1
513	329	1.4	518	35	0.1
520	311	1.3	505	33	0.1
512	292	1.2	517	25	0.1
506	257	1.1	516	23	0.1
515	221	0.9	522	198	0.8

More than 80% of the samples are in three shared spaces (nurse station, kitchen, and cafeteria), while some individual rooms have less than 50 samples. Such imbalance reflects realistic caregiving movement patterns and motivates the use of the weighted F1 score as the primary evaluation metric.

Table 4: Rooms exhibiting strong hour level concentration.

Room	Dominant Hour	Percentage (%)
510	10:00	100.0
505	11:00	87.9
522	15:00	86.9
515	10:00	82.4
520	10:00	74.9
501	14:00	76.6
503	11:00	70.5

This emphasis represents structured caring routines like scheduled visits and medication rounds. Similar hour distributions might show up in both training and testing sets under random stratified sampling, which might enable models to take use of temporal priors in addition to spatial signal patterns. If temporal overlap is not managed, this could result in optimistic estimations even though it does not amount to direct data leaking.

A.2 Leave-One-Day-Out Cross Validation (LODO-CV)

To evaluate robustness under day-level distribution shifts, Leave-One-Day-Out Cross Validation (LODO-CV) was applied. In this protocol, models are trained on three days and evaluated on a completely unseen fourth day, enforcing strict temporal isolation.

Table 5 presents pooled results across folds using the same hyperparameters as the main experiment. A downward ablation strategy was used, where one feature group is removed at a time while keeping all other components unchanged.

Table 5: Feature-group ablation under Leave-One-Day-Out cross validation (LODO-CV).

Configuration	**Accuracy**	**Macro F1**	**Weighted F1**
Baseline (All 84 Features)	0.5778	0.1889	0.5559
Without Basic RSSI Statistics	0.5769	0.1866	0.5547
Without Zone-based Features	0.5757	0.1891	0.5537
Without Inter-zone Ratios	0.5754	0.1878	0.5537
Without Proximity Transform	0.5757	0.1858	0.5539
Without Strongest-beacon Signature	0.5745	0.1813	0.5526
Without Location-specific Signals	0.5782	0.1890	0.5560
Without Temporal Indicators	0.5486	0.2069	0.5204

Performance under LODO-CV is lower than stratified sampling, reflecting inter-day distribution shifts and varying room frequencies across days.

Removing temporal indicators results in the largest reduction in weighted F1 (ΔWF1 ≈ -0.0355), indicating that temporal context contributes structured information even under strict day separation. Macro F1 increases slightly when temporal indicators are removed, suggesting redistribution of class-level performance under imbalance.

Removing spatial feature groups (zone aggregation, inter-zone ratios, proximity transform, strongest-beacon signature) leads to smaller but consistent decreases in weighted F1 (approximately -0.002 to -0.003), indicating complementary spatial encoding rather than reliance on a single abstraction. Basic RSSI statistics and location-specific signals show comparatively small marginal effects.

These observations indicate that spatial and temporal components operate jointly, with temporal context contributing substantially and spatial abstractions providing stable structural support across days.

B Model Selection and Hyperparameter Optimization

B.1 Impact of Feature Engineering

To assess the contribution of the engineered representation, XGBoost was evaluated using (i) raw RSSI inputs only and (ii) the full 84-dimensional feature set.

Table 6: Comparison of raw RSSI and engineered features (held-out test set).

Feature Set	Accuracy	Weighted F1
Raw RSSI	0.6384	0.5991
All Features	0.7458	0.7410

The engineered representation is associated with improved performance relative to raw RSSI inputs, suggesting that structured feature design contributes meaningfully beyond model selection alone. The tuned model was then evaluated on a held-out test set that was not used during tuning.

For reference, the tuned configuration provided modest improvements over the fixed-parameter setup (approximately +1.8% in both Accuracy and Weighted F1), indicating incremental gains from parameter optimization.

B.1.1 Overfitting Assessment

To assess potential overfitting, inner cross-validation performance was compared with held-out test performance:

- Inner CV Accuracy (5-fold): 0.7383
- Test Set Accuracy: 0.7458
- Gap (CV – Test): -0.0076

The difference between validation and test performance is below 1%, indicating no evidence of overfitting. The slightly higher test accuracy suggests stable generalization rather than memorization of training patterns.